Lectures 16–17 Stochastic gradient descent

Stochastic approximation · fluctuations and averaging

Reading guide. Section 1 is Lecture 16, Section 2 is Lecture 17. The tools are those of Lectures 12–15, cited as "Theorem 4.2 of Lectures 12–15" and so on: orthogonal increments, the L^2 convergence theorem, the martingale CLT, Freedman's inequality. Examples marked WORKED IN CLASS are done on the board. Exercises are typed A, B, C as before; the homework for a lecture is the exercise block closing its section.

1 Stochastic approximation

Stochastic gradient descent on a loss L with minimizer θ^* is the recursion $\theta_{k+1} = \theta_k - \eta_k\, g(\theta_k; z_{k+1})$, where $g(\theta_k; z_{k+1})$ is an unbiased but noisy estimate of $\nabla L(\theta_k)$ computed from a fresh sample. The two lectures prove two theorems about it, under strong convexity and smoothness of L and a variance bound on the noise.

- *A limit theorem* (Theorem 1.5). With step sizes $\eta_k = \eta/(k + k_0)$ and $\eta\mu > 1$,

$$\mathbb{E}\|\theta_n - \theta^*\|^2 \le \frac{C}{n},$$

  with an explicit C; with a constant step the error instead settles at a noise floor proportional to the step. This says how many samples a given accuracy costs: the mean-square error after n samples is of the same order as that of an estimator that uses all n samples at once, although each sample is looked at only once and then discarded.

- *A central limit theorem* (Theorems 2.2 and 2.4). At the scale $n^{-1/2}$ the error is Gaussian, $\sqrt{n}(\theta_n - \theta^*) \Rightarrow \mathcal{N}(0, \Sigma_\eta)$, with a covariance that depends on the step size η and is smallest at a value of η one cannot know in advance; averaging the iterates, $\bar\theta_n = \frac{1}{n}\sum_{k<n}\theta_k$, gives $\sqrt{n}(\bar\theta_n - \theta^*) \Rightarrow \mathcal{N}(0, H^{-1}\Sigma H^{-1})$, the smallest covariance any estimator based on n samples can have, for any choice of step. This is what a confidence interval for θ^* is built from, and it is the reason to average.

Both are consequences of the fact, from Lecture 12, that the noise in the recursion is a martingale difference: the limit theorem is the L^2 theory of Lectures 12–14 applied to that martingale, the central limit theorem is the martingale CLT of Lecture 14, and the trajectory bounds at the end of Lecture 17 are Freedman's inequality from Lecture 15.

1.1 Setting

Let $L : \mathbb{R}^d \to \mathbb{R}$ be continuously differentiable. We assume:

(A1) *Strong convexity.* $\langle \nabla L(\theta) - \nabla L(\theta'), \theta - \theta' \rangle \ge \mu\|\theta - \theta'\|^2$ for some $\mu > 0$.

(A2) *Smoothness.* $\|\nabla L(\theta) - \nabla L(\theta')\| \le \beta\|\theta - \theta'\|$ for some $\beta \ge \mu$.

Under (A1) L has exactly one minimizer θ^*, and it is the only point where ∇L vanishes. Indeed, with $\theta_t = \theta + t(\theta' - \theta)$, the fundamental theorem of calculus along the segment and then (A1) at the pair (θ_t, θ), for which $\theta_t - \theta = t(\theta' - \theta)$, give

$$L(\theta') - L(\theta) - \langle \nabla L(\theta), \theta' - \theta \rangle = \int_0^1 \langle \nabla L(\theta_t) - \nabla L(\theta), \theta' - \theta \rangle \, dt \;\geq\; \int_0^1 \mu t \, \|\theta' - \theta\|^2 \, dt = \tfrac{\mu}{2} \|\theta' - \theta\|^2,$$

so $L(\theta') \to \infty$ as $\|\theta'\| \to \infty$ and a minimizer exists (a continuous function on a large closed ball attains its minimum); at a minimizer the gradient is zero, and the same inequality with $\nabla L(\theta) = 0$ shows $L(\theta') > L(\theta)$ for every $\theta' \neq \theta$, so there is only one such point. Write $e = \theta - \theta^*$ for the error. Taking $\theta' = \theta^*$ in (A1)–(A2),

$$\langle \nabla L(\theta), e \rangle \geq \mu \|e\|^2, \qquad \|\nabla L(\theta)\| \leq \beta \|e\|. \tag{1}$$

The algorithm sees L only through stochastic gradients. Let $z_1, z_2, \ldots$ be i.i.d. samples and $g(\theta; z) \in \mathbb{R}^d$ a function of the parameter and the sample with

(A3) *Unbiased gradients with controlled variance.* $\mathbb{E} g(\theta; z) = \nabla L(\theta)$ and $\mathbb{E} \|g(\theta; z) - \nabla L(\theta)\|^2 \leq \sigma^2 + \kappa \|\theta - \theta^*\|^2$ for every θ, for constants $\sigma^2 \geq 0$, $\kappa \geq 0$.

The typical case is $L(\theta) = \mathbb{E}\, \ell(\theta; z)$ and $g = \nabla_\theta \ell$. The term $\kappa \|e\|^2$ allows the noise to grow with the distance from the minimizer, which it does in least squares (Example 1.6).

Stochastic gradient descent with step sizes $\eta_0, \eta_1, \cdots > 0$ is

$$\theta_{k+1} = \theta_k - \eta_k \, g(\theta_k; z_{k+1}), \qquad \theta_0 \text{ fixed.}$$

With $\mathcal{F}_k = (z_1, \ldots, z_k)$, $\theta_k \in L^2(\mathcal{F}_k)$ (coordinatewise; we assume throughout that the needed second moments are finite), and by the freezing lemma (Lemma 1.5 of Lectures 12–15) with $U = \theta_k$ and $Z = z_{k+1}$,

$$\xi_{k+1} := g(\theta_k; z_{k+1}) - \nabla L(\theta_k) \quad \text{satisfies} \quad \mathbb{E}_k \xi_{k+1} = 0, \qquad s^2(\theta_k) := \mathbb{E}_k \|\xi_{k+1}\|^2 \leq \sigma^2 + \kappa \|e_k\|^2.$$

So, as in Example 2.4(d) of Lectures 12–15,

$$e_{k+1} = e_k - \eta_k \nabla L(\theta_k) - \eta_k \xi_{k+1}, \qquad M_n := \sum_{k<n} \eta_k \xi_{k+1} \text{ a martingale with respect to } (\mathcal{F}_n). \tag{2}$$

1.2 The one-step recursion

Lemma 1.1 (one step)**.** *For every k,*

$$\mathbb{E}_k \|e_{k+1}\|^2 = \|e_k - \eta_k \nabla L(\theta_k)\|^2 + \eta_k^2 s^2(\theta_k) \;\leq\; \big(1 - 2\mu\eta_k + (\beta^2 + \kappa)\eta_k^2\big) \|e_k\|^2 + \eta_k^2 \sigma^2.$$

In particular, if $\eta_k \leq \mu/(\beta^2 + \kappa)$, then $\mathbb{E}_k \|e_{k+1}\|^2 \leq (1 - \mu\eta_k) \|e_k\|^2 + \eta_k^2 \sigma^2$.

Proof. Expand $\|e_{k+1}\|^2 = \|e_k - \eta_k \nabla L(\theta_k)\|^2 - 2\eta_k \langle e_k - \eta_k \nabla L(\theta_k), \xi_{k+1} \rangle + \eta_k^2 \|\xi_{k+1}\|^2$. The vector $e_k - \eta_k \nabla L(\theta_k)$ is in $L^2(\mathcal{F}_k)$, so by rule (iii) the cross term has $\mathbb{E}_k$ equal to zero, and $\mathbb{E}_k \|\xi_{k+1}\|^2 = s^2(\theta_k)$. For the inequality, by (1),

$$\|e_k - \eta_k \nabla L(\theta_k)\|^2 = \|e_k\|^2 - 2\eta_k \langle \nabla L(\theta_k), e_k \rangle + \eta_k^2 \|\nabla L(\theta_k)\|^2 \leq (1 - 2\mu\eta_k + \beta^2 \eta_k^2) \|e_k\|^2,$$

and (A3) bounds $\eta_k^2 s^2(\theta_k) \leq \eta_k^2 \sigma^2 + \eta_k^2 \kappa \|e_k\|^2$. The last claim uses $(\beta^2 + \kappa)\eta_k \leq \mu$. □

Taking expectations, $a_k := \mathbb{E}\|e_k\|^2$ satisfies, for $\eta_k \le \mu/(\beta^2+\kappa)$,

$$a_{k+1} \le (1-\mu\eta_k)\, a_k + \eta_k^2\sigma^2. \tag{3}$$

Unrolling (3) (induction on n) gives the explicit bound

$$a_n \;\le\; a_0 \prod_{k<n}(1-\mu\eta_k) \;+\; \sigma^2 \sum_{k<n} \eta_k^2 \prod_{j=k+1}^{n-1} (1-\mu\eta_j), \tag{4}$$

with the usual convention that an empty product is 1. The first term is the *bias*: the initial error, contracted by the factor $1-\mu\eta_k$ at each step, so by at most $\exp(-\mu\sum_{k<n}\eta_k)$ in total. The second is the *variance*: the noise $\eta_k^2\sigma^2$ injected at step k, contracted by every step after it. Everything in this lecture is read off (4).

1.3 Constant step size

Theorem 1.2 (constant step)**.** *If $\eta_k \equiv \eta \le \mu/(\beta^2+\kappa)$, then for every n,*

$$\mathbb{E}\|\theta_n - \theta^*\|^2 \le (1-\mu\eta)^n \|\theta_0-\theta^*\|^2 + \frac{\eta\sigma^2}{\mu}.$$

Proof. In (4) the products are powers of $1-\mu\eta$: $a_n \le (1-\mu\eta)^n a_0 + \eta^2\sigma^2\sum_{j<n}(1-\mu\eta)^j \le (1-\mu\eta)^n a_0 + \eta^2\sigma^2/(\mu\eta)$. □

The initial error is forgotten exponentially fast, at rate $\mu\eta$ per step, and the iterates then fluctuate at the *noise floor* $\eta\sigma^2/\mu$. Halving the step halves the floor and doubles the time to reach it; decreasing step sizes are the way to have both.

Remark 1.3 (the floor is real). Suppose, in addition to (A1)–(A3), that the noise does not vanish anywhere: $s^2(\theta) \ge \sigma_-^2 > 0$ for all θ. Then for a constant step $\eta \le \min(1/\beta, \mu/(\beta^2+\kappa))$,

$$\liminf_{n\to\infty} \mathbb{E}\|\theta_n-\theta^*\|^2 \;\ge\; \frac{\eta\,\sigma_-^2}{2\beta},$$

so Theorem 1.2 is sharp up to the factor $2\beta/\mu\cdot\sigma^2/\sigma_-^2$.

Derivation. Start from the identity in Lemma 1.1, before any inequality was used:

$$\mathbb{E}_k\|e_{k+1}\|^2 = \|e_k - \eta\nabla L(\theta_k)\|^2 + \eta^2 s^2(\theta_k).$$

The drift term from below. By the triangle inequality and (1), $\|e_k-\eta\nabla L(\theta_k)\| \ge \|e_k\|-\eta\|\nabla L(\theta_k)\| \ge (1-\beta\eta)\|e_k\|$, and the right side is ≥ 0 because $\beta\eta \le 1$; squaring, $\|e_k-\eta\nabla L(\theta_k)\|^2 \ge (1-\beta\eta)^2\|e_k\|^2$. *The noise term from below.* $\eta^2 s^2(\theta_k) \ge \eta^2\sigma_-^2$ by hypothesis. *Take expectations.* With $a_k = \mathbb{E}\|e_k\|^2$ and $q := (1-\beta\eta)^2 \in [0,1)$,

$$a_{k+1} \;\ge\; q\,a_k + \eta^2\sigma_-^2 \qquad \text{for all } k \ge 0.$$

Unroll. Induction on n (exactly as for (4), with the inequality reversed) gives

$$a_n \;\ge\; q^n a_0 + \eta^2\sigma_-^2 \sum_{j<n} q^j = q^n a_0 + \eta^2\sigma_-^2\,\frac{1-q^n}{1-q} \;\ge\; \eta^2\sigma_-^2\,\frac{1-q^n}{1-q}.$$

Let $n \to \infty$. Since $q < 1$, $q^n \to 0$ and $\liminf_n a_n \geq \eta^2\sigma_-^2/(1-q)$. Finally $1 - q = 1 - (1-\beta\eta)^2 = \beta\eta(2-\beta\eta) \leq 2\beta\eta$, so $\eta^2\sigma_-^2/(1-q) \geq \eta\sigma_-^2/(2\beta)$.

Exactness for a quadratic. For $L(\theta) = \frac{h}{2}(\theta - \theta^*)^2$ in $d = 1$ with noise of constant variance σ^2, both steps above are equalities: $\|e - \eta\nabla L\|^2 = (1-\eta h)^2 e^2$ and $s^2 \equiv \sigma^2$, so $a_{k+1} = (1-\eta h)^2 a_k + \eta^2\sigma^2$ and $a_n \to \eta^2\sigma^2/(1-(1-\eta h)^2) = \eta\sigma^2/(h(2-\eta h))$, which is Exercise 2.12(c) of Lectures 12–15 with $h = 1$. Here $\mu = \beta = h$ and the upper bound $\eta\sigma^2/\mu$ of Theorem 1.2 is off by the factor $2 - \eta h \in (1, 2]$.

When the floor is absent. The hypothesis $\sigma_- > 0$ cannot be dropped: if the noise vanishes at the minimizer, $s^2(\theta) \leq \kappa\|e\|^2$, then $a_{k+1} \leq (1-\mu\eta)a_k$ and $a_n \to 0$ at a constant step (Example 2.8).

1.4 Decreasing step sizes: the Robbins–Monro rate

Lemma 1.4 (a recursion)**.** *Let $a_{k+1} \leq \big(1 - \frac{c}{k+k_0}\big)a_k + \frac{C}{(k+k_0)^2}$ for all $k \geq 0$, with $c > 1$, $C \geq 0$ and $k_0 \geq 1$. Then*

$$a_k \leq \frac{D}{k+k_0} \quad \text{for all } k \geq 0, \qquad D := \max\Big(\frac{C}{c-1},\ k_0 a_0\Big).$$

Proof. Induction. The case $k = 0$ holds since $D \geq k_0 a_0$. If $a_k \leq D/m$ with $m = k + k_0$, then $a_{k+1} \leq \frac{D(m-c)+C}{m^2}$, and $\frac{D(m-c)+C}{m^2} \leq \frac{D}{m+1}$ is equivalent to $m\big(C - D(c-1)\big) \leq Dc - C$, which holds because the left side is ≤ 0 (as $D(c-1) \geq C$) and the right side is $\geq Dc - D(c-1) = D > 0$. □

Theorem 1.5 (Robbins–Monro)**.** *Let $\eta_k = \dfrac{\eta}{k+k_0}$ with $\eta\mu > 1$ and $k_0 \geq \eta(\beta^2+\kappa)/\mu$. Then for every n,*

$$\mathbb{E}\|\theta_n - \theta^*\|^2 \leq \frac{1}{n+k_0}\max\Big(\frac{\eta^2\sigma^2}{\eta\mu - 1},\ k_0\|\theta_0 - \theta^*\|^2\Big).$$

Proof. The choice of k_0 gives $\eta_k \leq \eta/k_0 \leq \mu/(\beta^2+\kappa)$ for all k, so (3) holds and reads $a_{k+1} \leq (1 - \frac{\eta\mu}{k+k_0})a_k + \frac{\eta^2\sigma^2}{(k+k_0)^2}$. Apply Lemma 1.4 with $c = \eta\mu$, $C = \eta^2\sigma^2$. □

The same rate can be read off (4) directly: with $m = k + k_0$ and $N = n + k_0$, $\prod_{j=k+1}^{n-1}(1-\mu\eta_j) \leq \exp(-\eta\mu\sum_{j=k+1}^{n-1}\frac{1}{j+k_0}) \leq \big(\frac{m+1}{N}\big)^{\eta\mu}$, so the bias is $O(N^{-\eta\mu})$ and the variance term is at most $\sigma^2\eta^2 N^{-\eta\mu}\sum_{m<N} m^{-2}(m+1)^{\eta\mu} = O(N^{-1})$ when $\eta\mu > 1$, since $\sum_{m<N} m^{\eta\mu-2} \asymp N^{\eta\mu-1}$. Lemma 1.4 is the same computation with the constant made explicit.

The mean-square error decays like $1/n$, the rate of an average of n independent observations, with constant $\eta^2\sigma^2/(\eta\mu - 1)$. Two features of the statement matter. The condition $\eta\mu > 1$ requires knowing a lower bound on the curvature μ: if $\eta\mu < 1$ the recursion only gives the slower rate $n^{-\eta\mu}$ (Exercise 1.9). And the constant is minimized at $\eta = 2/\mu$, not at the largest admissible step. Lecture 17 refines both: the sharp threshold is $\eta\mu > \frac{1}{2}$ when the recursion is linearized exactly, and averaging the iterates removes the need to know μ altogether.

Example 1.6 (least mean squares)**.** WORKED IN CLASS Linear regression with Gaussian design: $z = (x, y)$ with $x \sim \mathcal{N}(0, I_d)$, $y = \langle x, \theta^*\rangle + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ independent of x, and the loss $\ell(\theta; z) = \frac{1}{2}(\langle x, \theta\rangle - y)^2$. Then

$$L(\theta) = \tfrac{1}{2}\|\theta - \theta^*\|^2 + \tfrac{1}{2}\sigma_\varepsilon^2, \qquad \nabla L(\theta) = e, \qquad g(\theta; z) = x\,(\langle x, e\rangle - \varepsilon),$$

so $\mu = \beta = 1$ and SGD is the *least mean squares* algorithm $\theta_{k+1} = \theta_k - \eta_k x_{k+1}(\langle x_{k+1}, \theta_k\rangle - y_{k+1})$. The gradient noise is $\xi = x\langle x, e\rangle - e - x\varepsilon$, and by Wick's theorem ($\mathbb{E}[(x^\top e)^2\|x\|^2] = (d+2)\|e\|^2$, $\mathbb{E}\|x\|^2 = d$),

$$\mathbb{E}\|\xi\|^2 = \mathbb{E}\big[(x^\top e)^2\|x\|^2\big] - \|e\|^2 + \sigma_\varepsilon^2\mathbb{E}\|x\|^2 = (d+1)\|e\|^2 + d\sigma_\varepsilon^2.$$

So (A3) holds with $\kappa = d+1$ and $\sigma^2 = d\sigma_\varepsilon^2$, and nothing smaller: the noise is *multiplicative*, proportional to the current error, and its size is set by the dimension. Lemma 1.1 requires $\eta_k \le 1/(d+2)$; with $\eta_k = \eta/(k+k_0)$, $\eta > 1$, $k_0 \ge \eta(d+2)$, Theorem 1.5 gives

$$\mathbb{E}\|\theta_n - \theta^*\|^2 \le \frac{1}{n+k_0}\max\Big(\frac{\eta^2 d\,\sigma_\varepsilon^2}{\eta-1},\, k_0\|\theta_0-\theta^*\|^2\Big) \approx \frac{\eta^2}{\eta-1}\cdot\frac{d\sigma_\varepsilon^2}{n} \quad \text{for large } n,$$

which at $\eta = 2$ is $4d\sigma_\varepsilon^2/n$: four times the mean-square error $d\sigma_\varepsilon^2/n$ of the least squares estimator on the same n samples. Lecture 17 shows how to remove the factor 4.

1.5 The Robbins–Monro conditions

Proposition 1.7. *Suppose $\eta_k \le \mu/(\beta^2+\kappa)$ for all k, $\sum_k \eta_k = \infty$ and $\eta_k \to 0$. Then $\mathbb{E}\|\theta_n - \theta^*\|^2 \to 0$.*

Proof. From (3), $a_{k+1} \le (1-\mu\eta_k)a_k + \eta_k^2\sigma^2$. Given $\epsilon > 0$ choose n_0 with $\eta_k\sigma^2 \le \mu\epsilon$ for $k \ge n_0$; then $a_{k+1} - \epsilon \le (1-\mu\eta_k)(a_k - \epsilon)$ for $k \ge n_0$, so $(a_n - \epsilon)_+ \le (a_{n_0} - \epsilon)_+ \exp(-\mu\sum_{k=n_0}^{n-1}\eta_k) \to 0$. Hence $\limsup a_n \le \epsilon$. □

The conditions $\sum_k \eta_k = \infty$ and $\sum_k \eta_k^2 < \infty$ are Robbins and Monro's (1951). The first lets the drift reach θ^* from anywhere, and is all the mean-square statement needs. The second makes the accumulated noise $M_n = \sum_{k<n}\eta_k\xi_{k+1}$ a martingale that is bounded in L^2, hence convergent (Theorem 4.2 of Lectures 12–15); it is what controls the whole trajectory rather than a fixed time n, and Section 2.3 does this with Freedman's inequality.

1.6 The ODE picture

Write $t_k = \sum_{j<k}\eta_j$ for the accumulated step size. Then (2) is

$$\theta_{k+1} - \theta_k = -(t_{k+1} - t_k)\,\nabla L(\theta_k) - \eta_k\xi_{k+1}\,:$$

Euler's scheme for the gradient flow $\dot\theta = -\nabla L(\theta)$ with time steps η_k, perturbed by a martingale whose quadratic variation up to time t_k is $\sum_{j<k}\eta_j^2 s^2(\theta_j)$. Under the Robbins–Monro conditions the ODE time t_k goes to infinity while the noise has finite total quadratic variation: on the ODE's time scale the noise is negligible. For strongly convex L the flow satisfies $\|e(t)\| \le e^{-\mu t}\|e(0)\|$; with $\eta_k = \eta/k$, $t_k \approx \eta\log k$, so the bias decays like $k^{-\mu\eta}$, which is the rate $n^{-\eta\mu}$ seen for $\eta\mu < 1$ and the reason the $1/n$ rate needs $\eta\mu > 1$: for the bias to be forgotten faster than the variance accumulates, the ODE must contract faster than $1/\sqrt{k}$ in k.

Exercises for Lecture 16

Exercise 1.8. TYPE A Let $L(\theta) = \frac{1}{2}\theta^\top H\theta$ with H symmetric, $\mu I \preceq H \preceq \beta I$. Verify (A1)–(A2) and show $\nabla L(\theta) = H\theta$. For $g(\theta; z) = H\theta + z$ with z centered of covariance Σ, verify (A3) with $\kappa = 0$, $\sigma^2 = \operatorname{tr}\Sigma$, and compute $\mathbb{E}\|e_n\|^2$ exactly for constant step in the case $H = hI$.

Exercise 1.9. type B Let $a_{k+1} \le (1-\frac{c}{k+k_0})a_k + \frac{C}{(k+k_0)^2}$ with $0<c<1$. Show $a_k \le D(k+k_0)^{-c}$ for a constant D depending on a_0, C, c, k_0 (try $D(k+k_0)^{-c}$ in the induction). Conclude that for $\eta_k = \eta/(k+k_0)$ with $\eta\mu < 1$, Theorem 1.5 is replaced by the rate $n^{-\eta\mu}$. Show by the quadratic of Exercise 1.8 in $d=1$ that this rate is attained.

Exercise 1.10 (mini-batches)**.** type B Replace $g(\theta;z)$ by the average of B independent stochastic gradients. Show (A3) holds with (σ^2,κ) replaced by $(\sigma^2/B, \kappa/B)$, and derive the corresponding versions of Theorems 1.2 and 1.5. In Theorem 1.2, compare, for a fixed budget of nB gradient evaluations, the floor reached by batch size B with step η against batch size 1 with step η/B.

Exercise 1.11 (step size $\eta_k \propto k^{-\alpha}$)**.** type B Let $\eta_k = \eta(k+k_0)^{-\alpha}$ with $\alpha \in (\frac12, 1)$ and $\eta_k \le \mu/(\beta^2+\kappa)$. Show from (3) that $a_k \le D\,\eta_k$ for a constant D and all k, provided k_0 is large enough (induction with $D \ge \max(2\sigma^2/\mu, a_0/\eta_0)$, using $\eta_{k+1}/\eta_k \ge 1-\alpha/(k+k_0)$ and $2\alpha(k+k_0)^{\alpha-1} \le \eta\mu$), hence $\mathbb{E}\|e_k\|^2 = O(k^{-\alpha})$. This is the rate Lecture 17 needs for averaging.

Exercise 1.12 (non-convex smooth losses)**.** type C Drop (A1) and assume only (A2), $L \ge L_*$, and (A3) with $\kappa = 0$. (a) Use the descent inequality $L(\theta') \le L(\theta) + \langle \nabla L(\theta), \theta'-\theta\rangle + \frac{\beta}{2}\|\theta'-\theta\|^2$ to show $\mathbb{E}_k L(\theta_{k+1}) \le L(\theta_k) - \eta_k(1-\frac{\beta\eta_k}{2})\|\nabla L(\theta_k)\|^2 + \frac{\beta}{2}\eta_k^2\sigma^2$. (b) Sum, for $\eta_k \le 1/\beta$, to obtain

$$\min_{k<n} \mathbb{E}\|\nabla L(\theta_k)\|^2 \;\le\; \frac{2\big(L(\theta_0)-L_*\big) + \beta\sigma^2\sum_{k<n}\eta_k^2}{\sum_{k<n}\eta_k}.$$

(c) Optimize over a constant step for a horizon n, and over $\eta_k \propto k^{-1/2}$. What does the bound say, and what does it not say, about where the iterates end up?

Exercise 1.13 (LMS, simulated)**.** type C Implement Example 1.6 with $d=20$, $\sigma_\varepsilon = 1$, θ^* a fixed unit vector, $\theta_0 = 0$. (a) Run $\eta_k = \eta/(k+k_0)$ for $\eta \in \{0.5, 1, 2, 4\}$ with $k_0 = \eta(d+2)$, $n = 10^4$, 200 repetitions, and plot $n\,\mathbb{E}\|\theta_n - \theta^*\|^2$ against η; compare with $\eta^2 d/(\eta-1)$ and with the least squares value d. (b) Run the constant step $\eta = 1/(d+2)$ and locate the noise floor. (c) Run $\eta_k = \eta k^{-0.7}$ with the averaged iterate $\bar\theta_n = \frac1n\sum_{k<n}\theta_k$ and compare $n\mathbb{E}\|\bar\theta_n - \theta^*\|^2$ with d.

2 Fluctuations and averaging

Lecture 16 bounded the mean-square error. This lecture identifies the distribution of the error at scale $n^{-1/2}$, and shows that averaging the iterates attains the smallest possible limiting variance without knowledge of the curvature. To keep the notation light the proofs are written for $d=1$; Remark 2.5 states the vector versions. In addition to (A1)–(A3) we assume:

(A4) *Linearization.* $L'(\theta) = h\,e + r(\theta)$ with $h = L''(\theta^*) \ge \mu$ and $|r(\theta)| \le \frac{\gamma}{2}e^2$.

(A5) *Noise regularity.* $\mathbb{E}_k|\xi_{k+1}|^3 \le \rho^3$ for all k, and the conditional variance $s^2(\theta) = \mathbb{E}|g(\theta;z) - L'(\theta)|^2$ is continuous at θ^*, with $\sigma_*^2 := s^2(\theta^*) > 0$.

With (A4) the recursion (2) becomes

$$e_{k+1} = (1-\eta_k h)\,e_k - \eta_k\xi_{k+1} - \eta_k r_k, \qquad r_k := r(\theta_k), \quad |r_k| \le \tfrac{\gamma}{2}e_k^2. \tag{5}$$

Lemma 2.1. *If* $\mathbb{E}e_k^2 \to 0$ *then* $\mathbb{E}|s^2(\theta_k) - \sigma_*^2| \to 0$.

Proof. Given $\epsilon > 0$ pick δ with $|s^2(\theta) - \sigma_*^2| \le \epsilon$ for $|e| \le \delta$. On $\{|e_k| > \delta\}$ use $s^2(\theta_k) \le \sigma^2 + \kappa e_k^2$: $\mathbb{E}|s^2(\theta_k) - \sigma_*^2| \le \epsilon + (\sigma^2+\sigma_*^2)\mathbb{P}(|e_k|>\delta) + \kappa\mathbb{E}e_k^2 \le \epsilon + (\sigma^2+\sigma_*^2)\mathbb{E}e_k^2/\delta^2 + \kappa\mathbb{E}e_k^2$. □

2.1 Asymptotic normality of the iterates

Theorem 2.2 (asymptotic normality)**.** *Let $\eta_k = \eta/(k+k_0)$ with $\eta\mu > 1$ and $k_0 \geq \max\big(\eta(\beta^2+\kappa)/\mu,\, 2\eta h\big)$, and assume (A1)–(A5). Then*

$$\sqrt{n}\,(\theta_n - \theta^*) \;\Rightarrow\; \mathcal{N}\Big(0,\; \frac{\eta^2\sigma_*^2}{2\eta h - 1}\Big).$$

The limiting variance is the one found for the quadratic in Exercise 4.18 of Lectures 12–15, with σ_*^2, the noise variance *at the minimizer*, in place of σ^2. It is minimized at $\eta = 1/h$, where it equals σ_*^2/h^2; for the step $\eta = 2/\mu$ suggested by Theorem 1.5 it is $\frac{\sigma_*^2}{h^2}\cdot\frac{4r^2}{4r-1}$ with $r = h/\mu \geq 1$, larger by a factor of at least $\frac{4}{3}$. Choosing η well requires knowing h.

The proof unrolls (5). With

$$P_{n,k} := \prod_{j=k}^{n-1}(1-\eta_j h)\;\; (P_{n,n}=1), \qquad w_{n,k} := \eta_k P_{n,k+1},$$

$$e_n = P_{n,0}\,e_0 - \sum_{k<n} w_{n,k}\,\xi_{k+1} - \sum_{k<n} w_{n,k}\,r_k. \tag{6}$$

Lemma 2.3 (the weights)**.** *Under the hypotheses of Theorem 2.2, with $m = k+k_0$ and $N = n+k_0$: $0 \leq P_{n,k} \leq (m/N)^{\eta h}$; $w_{n,k} \leq \frac{\eta}{m}(m/N)^{\eta h}$; and*

$$n\sum_{k<n} w_{n,k}^2 \to \frac{\eta^2}{2\eta h-1}, \qquad n\sum_{k<K} w_{n,k}^2 \to 0 \;\; \textit{for each fixed } K, \qquad \sqrt{n}\sum_{k<n}\frac{w_{n,k}}{m} = O(n^{-1/2}).$$

Proof. $k_0 \geq 2\eta h$ gives $\eta_j h \leq \frac{1}{2}$, so $0 \leq 1-\eta_j h \leq e^{-\eta_j h}$ and $P_{n,k} \leq \exp(-h\sum_{j=k}^{n-1}\eta_j) \leq \exp(-\eta h\log(N/m))$, using $\sum_{j=k}^{n-1}\frac{1}{j+k_0} \geq \log\frac{N}{m}$. For the lower bound, $\log(1-x) \geq -x-x^2$ on $[0,\frac{1}{2}]$ gives $P_{n,k} \geq (m/N)^{\eta h}e^{-\eta h/m-\eta^2h^2/(m-1)}$, so $w_{n,k} = \frac{\eta}{m}(m/N)^{\eta h}(1+O(1/m))$. Then $n\sum_{k<n}w_{n,k}^2 = (1+o(1))\,\eta^2N^{1-2\eta h}\sum_{m=k_0}^{N-1} m^{2\eta h-2} \to \eta^2/(2\eta h-1)$, since $2\eta h-2 > -1$ and the terms with $m \leq K+k_0$ contribute $O(N^{1-2\eta h}) \to 0$. Finally $\sqrt{n}\sum_k w_{n,k}/m \leq \sqrt{n}\,\eta N^{-\eta h}\sum_m m^{\eta h-2} = O(\sqrt{n}\,N^{-\eta h}N^{\eta h-1})$ because $\eta h > 1$. □

Proof of Theorem 2.2. Multiply (6) by $\sqrt{n}$ and treat the three terms.

Initial condition. $\sqrt{n}\,P_{n,0}|e_0| \leq \sqrt{n}\,(k_0/N)^{\eta h}|e_0| \to 0$ since $\eta h > \frac{1}{2}$.

Remainder. By Theorem 1.5, $\mathbb{E}e_k^2 \leq D/m$, so $\mathbb{E}\big|\sqrt{n}\sum_{k<n}w_{n,k}r_k\big| \leq \frac{\gamma D}{2}\sqrt{n}\sum_{k<n}\frac{w_{n,k}}{m} = O(n^{-1/2})$ by Lemma 2.3: this term tends to 0 in L^1, hence in probability.

Martingale term. The differences $\zeta_{n,k+1} := \sqrt{n}\,w_{n,k}\xi_{k+1}$, $k<n$, are martingale differences (bounded predictable weights times ξ_{k+1}) with conditional variances $n\,w_{n,k}^2 s^2(\theta_k)$ and conditional third moments at most $n^{3/2}w_{n,k}^3\rho^3$. They are not independent: the law of ξ_{k+1} and its conditional variance $s^2(\theta_k)$ depend on θ_k, hence on $z_1,\dots,z_k$ (in Example 1.6, $\xi = xx^\top e - e - x\varepsilon$). So this is the martingale CLT and not Lindeberg's theorem; the two coincide only for additive noise $g(\theta;z) = \nabla L(\theta) + z$ with z independent of θ, as in Exercise 4.18 of Lectures 12–15. Check the hypotheses of Corollary 4.6 of Lectures 12–15 after dividing by $\tau^2 := \eta^2\sigma_*^2/(2\eta h-1)$:

- *Quadratic variation.* $V_n = \tau^{-2} n \sum_{k<n} w_{n,k}^2 s^2(\theta_k)$. Split at a fixed K: the terms $k < K$ contribute $O\big(n \sum_{k<K} w_{n,k}^2\big) \to 0$ in L^1 by Lemma 2.3 and $s^2 \le \sigma^2 + \kappa e_k^2$; for $k \ge K$, $\mathbb{E}\big|n \sum_{k\ge K} w_{n,k}^2 (s^2(\theta_k) - \sigma_*^2)\big| \le \sup_{k\ge K} \mathbb{E}|s^2(\theta_k) - \sigma_*^2| \cdot n \sum_k w_{n,k}^2$, which is small for K large by Lemma 2.1 (as $\mathbb{E}e_k^2 \to 0$). With $n \sum_k w_{n,k}^2 \to \eta^2/(2\eta h - 1)$ this gives $\mathbb{E}|V_n - 1| \to 0$.
- *Third moments.* $\sum_{k<n} \mathbb{E}|\zeta_{n,k+1}|^3 \le \rho^3 n^{3/2} \sum_k w_{n,k}^3 \le \rho^3 \eta^3 n^{3/2} N^{-3\eta h} \sum_m m^{3\eta h - 3} = O(n^{-1/2})$, as $3\eta h - 3 > -1$.
- *Largest variance.* $\max_k n\, w_{n,k}^2 s^2(\theta_k) \le n \max_k w_{n,k}^2 \cdot (\sigma^2 + \kappa \max_{k<n} e_k^2)$, and $n \max_k w_{n,k}^2 = O(1/n)$ (the bound $\frac{\eta}{m}(m/N)^{\eta h}$ is increasing in m for $\eta h > 1$), while $\mathbb{E} \max_{k<n} e_k^2 \le \sum_{k<n} \mathbb{E}e_k^2 = O(\log n)$; the product tends to 0.

Hence $\tau^{-1}\sqrt{n} \sum_{k<n} w_{n,k} \xi_{k+1} \Rightarrow \mathcal{N}(0,1)$. The three terms combine by Slutsky's lemma (a sum of a sequence converging in distribution and sequences converging to 0 in probability converges in distribution to the same limit). □

2.2 Polyak–Ruppert averaging

The optimal variance σ_*^2/h^2 in Theorem 2.2 needs $\eta = 1/h$. Averaging the iterates obtains it with no knowledge of h, and with a larger step size, which also forgets the initial condition faster.

Theorem 2.4 (Polyak–Ruppert)**.** *Let $\eta_k = \eta(k + k_0)^{-\alpha}$ with $\alpha \in (\frac{1}{2}, 1)$ and $\eta_k \le \mu/(\beta^2 + \kappa)$, let (A1)–(A5) hold, and let $\bar\theta_n = \frac{1}{n}\sum_{k<n} \theta_k$. Then*

$$\sqrt{n}\,(\bar\theta_n - \theta^*) \;\Rightarrow\; \mathcal{N}\Big(0,\; \frac{\sigma_*^2}{h^2}\Big).$$

Proof. By Exercise 1.11 (with k_0 large enough), $\mathbb{E}e_k^2 \le D\eta_k = O(k^{-\alpha})$. Solve (5) for he_k and average:

$$h\,\bar e_n = \frac{1}{n}\sum_{k<n} \frac{e_k - e_{k+1}}{\eta_k} \;-\; \frac{1}{n}\sum_{k<n} \xi_{k+1} \;-\; \frac{1}{n}\sum_{k<n} r_k, \qquad \bar e_n := \bar\theta_n - \theta^*.$$

Multiply by $\sqrt{n}$.

Martingale term. $\frac{1}{\sqrt{n}}\sum_{k<n} \xi_{k+1}$ has differences $\xi_{k+1}/\sqrt{n}$ with conditional variances $s^2(\theta_k)/n$; as in Theorem 2.2, these are dependent, with a random conditional variance, so the martingale CLT is the tool. Its quadratic variation $\frac{1}{n}\sum_{k<n} s^2(\theta_k)$ tends to σ_*^2 in L^1 by Lemma 2.1 (Cesàro); the third moments sum to at most $n\rho^3 n^{-3/2} \to 0$; and the largest conditional variance is at most $(\sigma^2 + \kappa \max_{k<n} e_k^2)/n$ with $\mathbb{E}\max_{k<n} e_k^2 \le \sum_{k<n} D\eta_k = O(n^{1-\alpha})$, so it tends to 0 in L^1. By Corollary 4.6 of Lectures 12–15, $\frac{1}{\sqrt{n}}\sum_{k<n} \xi_{k+1} \Rightarrow \mathcal{N}(0, \sigma_*^2)$.

Telescoping term. Summation by parts,

$$\sum_{k<n} \frac{e_k - e_{k+1}}{\eta_k} = \frac{e_0}{\eta_0} - \frac{e_n}{\eta_{n-1}} + \sum_{k=1}^{n-1} e_k\Big(\frac{1}{\eta_k} - \frac{1}{\eta_{k-1}}\Big),$$

and $0 \le \frac{1}{\eta_k} - \frac{1}{\eta_{k-1}} \le \frac{\alpha}{\eta}(k+k_0)^{\alpha-1}$. With $\mathbb{E}|e_k| \le \sqrt{D\eta_k} = O(k^{-\alpha/2})$: $\mathbb{E}\frac{|e_n|}{\sqrt{n}\,\eta_{n-1}} = O(n^{-1/2+\alpha-\alpha/2}) = O(n^{(\alpha-1)/2}) \to 0$, and $\frac{1}{\sqrt{n}}\sum_{k<n} k^{\alpha-1} k^{-\alpha/2} = O(n^{-1/2} n^{\alpha/2}) \to 0$. So the telescoping term tends to 0 in L^1.

Remainder. $\mathbb{E}\frac{1}{\sqrt{n}}\sum_{k<n} |r_k| \le \frac{\gamma}{2\sqrt{n}} \sum_{k<n} D\eta_k = O(n^{1/2-\alpha}) \to 0$ because $\alpha > \frac{1}{2}$.

Slutsky's lemma gives $\sqrt{n}\,h\bar{e}_n \Rightarrow \mathcal{N}(0,\sigma_*^2)$. □

Both constraints on α are visible in the proof: $\alpha < 1$ makes the iterates move slowly enough that the telescoping term is negligible, $\alpha > \frac{1}{2}$ makes $\mathbb{E}e_k^2$ decay fast enough that the curvature error is negligible. The averaged iterate has, to first order, the fluctuations of $\frac{1}{n}\sum \xi_{k+1}$ divided by h: it behaves like the estimator that solves the population equation $L'(\theta) = 0$ with the gradient noise averaged out, which is the statistically efficient variance σ_*^2/h^2. For least mean squares (Example 1.6) this is $d\sigma_\varepsilon^2/n$ in mean square, the least squares value.

Remark 2.5 (vectors). For $d > 1$ let $H = \nabla^2 L(\theta^*)$ and $\Sigma = \mathbb{E}_{\theta^*}[\xi\xi^\top]$ the noise covariance at the minimizer. Theorem 2.2 becomes $\sqrt{n}(\theta_n - \theta^*) \Rightarrow \mathcal{N}(0,\Sigma_\eta)$ with Σ_η the solution of the Lyapunov equation $(\eta H - \frac{1}{2}I)\Sigma_\eta + \Sigma_\eta(\eta H - \frac{1}{2}I) = \eta^2\Sigma$, which requires $\eta\lambda_{\min}(H) > \frac{1}{2}$; Theorem 2.4 becomes $\sqrt{n}(\bar{\theta}_n - \theta^*) \Rightarrow \mathcal{N}(0, H^{-1}\Sigma H^{-1})$. The proofs are the same applied to $\langle v, e_n\rangle$ for each fixed $v \in \mathbb{R}^d$ (Cramér–Wold), with products of matrices in place of products of scalars; the weights do not commute in general, which is the only new bookkeeping.

2.3 High-probability bounds along the trajectory

Theorems 1.5 and 2.2 describe a fixed time n. Freedman's inequality controls the whole trajectory at once, through the noise martingale $M_n = \sum_{k<n} \eta_k \xi_{k+1}$ and its realized quadratic variation.

Proposition 2.6. *Let $L(\theta) = \frac{h}{2}(\theta - \theta^*)^2$ (so $r \equiv 0$), let the steps be nonincreasing with $\eta_k h \le 1$, and let $|\xi_k| \le c$. Then for every n,*

$$\big|e_n - P_{n,0}e_0\big| \;\le\; 2\max_{k\le n}|M_k|,$$

and consequently, for all N and $u, v > 0$,

$$\mathbb{P}\Big(\exists\, n \le N:\ |e_n - P_{n,0}e_0| \ge 2u \quad \text{and} \quad \sum_{k<n}\eta_k^2 s^2(\theta_k) \le v\Big) \;\le\; 2\exp\Big(-\frac{u^2}{2(v + \eta_0 c u/3)}\Big).$$

Proof. By (6) with $r \equiv 0$, $e_n - P_{n,0}e_0 = -\sum_{k<n} P_{n,k+1}(M_{k+1} - M_k)$. Summation by parts, with $P_{n,n} = 1$, $M_0 = 0$ and $P_{n,k+1} - P_{n,k} = \eta_k h P_{n,k+1} \ge 0$:

$$\sum_{k<n} P_{n,k+1}(M_{k+1} - M_k) = M_n - \sum_{k=1}^{n-1}\big(P_{n,k+1} - P_{n,k}\big)M_k, \qquad \sum_{k=1}^{n-1}\big(P_{n,k+1} - P_{n,k}\big) = 1 - P_{n,1} \le 1.$$

So the sum is bounded by $|M_n| + \max_{k<n}|M_k|$. For the probability bound, the steps of M are at most $\eta_0 c$ and $\langle M\rangle_n = \sum_{k<n}\eta_k^2 s^2(\theta_k)$ is nondecreasing, so on the event some $k \le n$ has $|M_k| \ge u$ and $\langle M\rangle_k \le v$; apply Theorem 5.10 of Lectures 12–15 to M and to $-M$. □

The bound is uniform over the horizon and sees the variance actually accumulated. Two consequences.

Example 2.7 (tail of the trajectory)**.** With $\eta_k = \eta/(k + k_0)$ and $s^2 \le \sigma^2$, apply the proposition to the shifted martingale $M_n - M_m$: for $n \ge m$ the noise contribution after time m is at most $2\max_{m\le k\le n}|M_k - M_m|$, and $\sum_{k\ge m}\eta_k^2\sigma^2 \le \eta^2\sigma^2/(m + k_0 - 1)$. So with probability at least $1 - \delta$, simultaneously for all $n \ge m$, the noise contributes at most $2\eta\sigma\sqrt{2\log(2/\delta)/(m + k_0 - 1)}\,(1 + o(1))$: the $1/\sqrt{m}$ rate of Theorem 1.5, now for the whole future at once, with a $\sqrt{\log(1/\delta)}$ price.

Example 2.8 (interpolation). WORKED IN CLASS If the noise vanishes at the minimizer, $s^2(\theta) \le \kappa e^2$ (so $\sigma = 0$: every sample gradient is zero at θ^*, as for a model that fits all the data), then Lemma 1.1 has no additive term: $a_{k+1} \le (1 - \mu\eta_k)a_k$ for $\eta_k \le \mu/(\beta^2 + \kappa)$. A constant step gives $\mathbb{E}\|e_n\|^2 \le (1 - \mu\eta)^n \|e_0\|^2$ with no noise floor, and $\eta_k = \eta/(k + k_0)$ gives $O(n^{-\eta\mu})$, faster than $1/n$ when $\eta\mu > 1$. In Proposition 2.6 the realized variance $\sum_{k<n} \eta_k^2 s^2(\theta_k) \le \kappa \sum_{k<n} \eta_k^2 e_k^2$ is then itself small along the trajectory, and Freedman's bound improves with it, while Azuma's, which only knows $|\xi_k| \le c$, does not (Exercise 5.17 of Lectures 12–15).

Exercises for Lecture 17

Exercise 2.9. TYPE A For $H = \operatorname{diag}(h_1, \dots, h_d)$ and $\Sigma = \operatorname{diag}(s_1^2, \dots, s_d^2)$ solve the Lyapunov equation of Remark 2.5 and show $\operatorname{tr} \Sigma_\eta = \sum_i \eta^2 s_i^2/(2\eta h_i - 1)$. Which η minimizes it when all h_i are equal? When they are not, compare $\min_\eta \operatorname{tr} \Sigma_\eta$ with $\operatorname{tr}(H^{-1}\Sigma H^{-1})$ for $d = 2$, $h = (1, 10)$, $s \equiv 1$.

Exercise 2.10. TYPE B In the linear case $L(\theta) = \frac{h}{2}\theta^2$ with i.i.d. noise of variance σ^2 and $\eta_k = \eta/(k + k_0)$, show that $n\,\mathbb{E}e_n^2 \to \infty$ when $\eta h < \frac{1}{2}$ and $n\mathbb{E}e_n^2 \asymp \log n$ when $\eta h = \frac{1}{2}$ (compute $n\sum_k w_{n,k}^2$ as in Lemma 2.3). So the threshold $\eta h > \frac{1}{2}$ in Theorem 2.2 is sharp, and Theorem 1.5's $\eta\mu > 1$ is an artefact of bounding $(1 - \eta_k h)^2$ by $1 - \eta_k h$.

Exercise 2.11 (tail averaging). TYPE B Let $\bar\theta_n^{(1/2)} = \frac{2}{n}\sum_{n/2 \le k < n} \theta_k$ (average of the second half). Adapt the proof of Theorem 2.4 to show $\sqrt{n}(\bar\theta_n^{(1/2)} - \theta^*) \Rightarrow \mathcal{N}(0, 2\sigma_*^2/h^2)$, and explain in one paragraph why one might accept the factor 2 (hint: how do $\bar\theta_n$ and $\bar\theta_n^{(1/2)}$ depend on θ_0 for moderate n?).

Exercise 2.12 (constant step plus averaging). TYPE B For the linear case with constant step η ($\eta h < 1$) and i.i.d. noise, compute $\lim_n n \operatorname{Var} \bar\theta_n$ exactly, and show it equals σ^2/h^2: with a constant step the averaged iterate is still efficient, although the last iterate is not (Theorem 1.2). Where did the step size go?

Exercise 2.13 (averaging, simulated). TYPE C For least mean squares (Example 1.6) with $d = 20$, compare over 200 runs and $n = 10^4$: the last iterate with $\eta_k = 2/(k + k_0)$; the last iterate with $\eta_k = 1/(k + k_0)$ (the optimal $\eta = 1/h$); the averaged iterate with $\eta_k = \eta k^{-0.7}$ for $\eta \in \{0.2, 0.5, 1\}/(d + 2)$. Report $n\mathbb{E}\|\cdot - \theta^*\|^2$ for each and compare with $4d$, d and d respectively. Then repeat with a badly conditioned design $x \sim \mathcal{N}(0, \operatorname{diag}(1, \dots, 1, 0.05))$ and comment on which methods needed to know the conditioning.

Exercise 2.14 (the vector case). TYPE C Write out the proof of Theorem 2.4 for $d > 1$: start from $H\bar e_n = \frac{1}{n}\sum(e_k - e_{k+1})/\eta_k - \frac{1}{n}\sum \xi_{k+1} - \frac{1}{n}\sum r_k$, apply the one-dimensional argument to $\langle v, \cdot \rangle$ for a fixed v, and conclude with the Cramér–Wold device (quote it). Identify the covariance of $\frac{1}{\sqrt{n}}\sum \xi_{k+1}$ and deduce the limit $\mathcal{N}(0, H^{-1}\Sigma H^{-1})$.